

Lab Worksheets – 2307421L- Ubiquitous Sensing, Computing and Communication (USCC)

Practical NO. 3: Cloud Connectivity & Real-Time Database

Batch: Batch

Group No.:

Members:

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Objective:

- To establish cloud connectivity between ESP8266 and Firebase Realtime Database. Configure and use a real-time cloud database.
- To upload multiple sensor readings (DHT11, Soil Moisture, and pH) in real-time.
- To retrieve and visualize the stored data for monitoring and analysis.
- To understand Firebase authentication and IoT data structuring.

Tools & Technology Used:

Tool / Component	Purpose
ESP8266	Microcontroller for data sensing and Wi-Fi communication
DHT11 Sensor	Measures temperature and humidity
Soil Moisture Sensor	Measures soil water content
pH Sensor	Monitors soil acidity/alkalinity
Arduino IDE	Coding and uploading firmware
Firebase	Cloud-based data logging
USB Cable & Jumper Wires	Hardware connections

Procedure:

- Configured Firebase and copied Database URL and secret key.
- Connected ESP8266 to local Wi-Fi network.
- Wrote Arduino code to read sensor values and send them to Firebase.
- Uploaded program to ESP8266 and verified data transmission in Serial Monitor.
- Checked Firebase for live JSON updates corresponding to sensor readings.



Code Snippets / Configuration:

```
#include <ESP8266WiFi.h>
#include <Firebase_ESP_Client.h>
#include <DHT.h>

// ----- WiFi & Firebase Credentials -----
#define WIFI_SSID "Airtel_Mighty Raju"
#define WIFI_PASSWORD "Mitaoe#701"

#define API_KEY "AIzaSyAUxLwgUALpPcWpp7_5GwHyNHsgNJQ0308" // your full key
#define DATABASE_URL "https://smart-agriculture-df146-default-rtdb.firebaseio.com/"

// ----- Sensor Pins -----
#define DHTPIN D4
#define DHTTYPE DHT11
#define ANALOG_PIN A0

DHT dht(DHTPIN, DHTTYPE);

FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig config;

// ----- Setup -----
void setup() {
  Serial.begin(115200);
  delay(1000);
```

```

WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
Serial.print("Connecting to WiFi");
while (WiFi.status() != WL_CONNECTED) {
  Serial.print(".");
  delay(500);
}
Serial.println("\n✔WiFi connected!");

// Firebase configuration
config.api_key = API_KEY;
config.database_url = DATABASE_URL;

// Anonymous sign-in
if (Firebase.signUp(&config, &auth, "", "")) {
  Serial.println("✔Firebase sign-up successful!");
} else {
  Serial.printf("✗Sign-up Error: %s\n", config.signer.signupError.message.c_str());
}

Firebase.begin(&config, &auth);
Firebase.reconnectWiFi(true);

dht.begin();
Serial.println("✔Firebase initialized!");
}

// ----- Sensor Functions -----
float readPH() {
  int samples = 10;
  float total = 0;
  for (int i = 0; i < samples; i++) {
    total += analogRead(ANALOG_PIN);
    delay(10);
  }
  float avgADC = total / samples;
  float voltage = avgADC * (3.3 / 1023.0);
  float pHValue = 7 + ((2.5 - voltage) / 0.18);
  return pHValue;
}

int readMoisture() {
  int val = analogRead(ANALOG_PIN);
  return map(val, 1023, 0, 0, 100);
}

// ----- Main Loop -----
void loop() {
  float temperature = dht.readTemperature();

```

```

float humidity = dht.readHumidity();
int moisture = readMoisture();
float pH = readPH();

Serial.println("\n  Sensor Readings:");
Serial.printf("Temperature: %.2f °C\n", temperature);
Serial.printf("Humidity: %.2f %%\n", humidity);
Serial.printf("Moisture: %d %%\n", moisture);
Serial.printf("pH Value: %.2f\n", pH);

if (Firebase.ready()) {
  // Create unique path using timestamp
  String path = "/SmartFarm/Readings/" + String(millis());

  Firebase.RTDB.setFloat(&fbdo, path + "/SmartAgri/Temperature", temperature);
  Firebase.RTDB.setFloat(&fbdo, path + "/SmartAgri/Humidity", humidity);
  Firebase.RTDB.setInt(&fbdo, path + "/SmartAgri/Moisture", moisture);
  Firebase.RTDB.setFloat(&fbdo, path + "/SmartAgri/pH", pH);

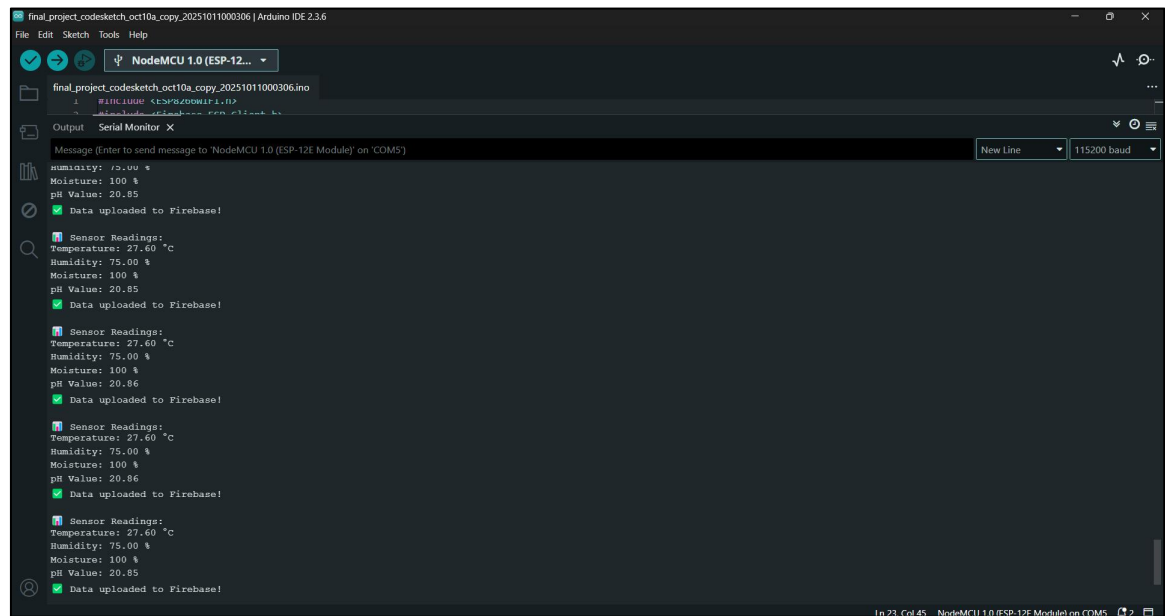
  Serial.println("✔ Data uploaded to Firebase!");
}

delay(10000);
}

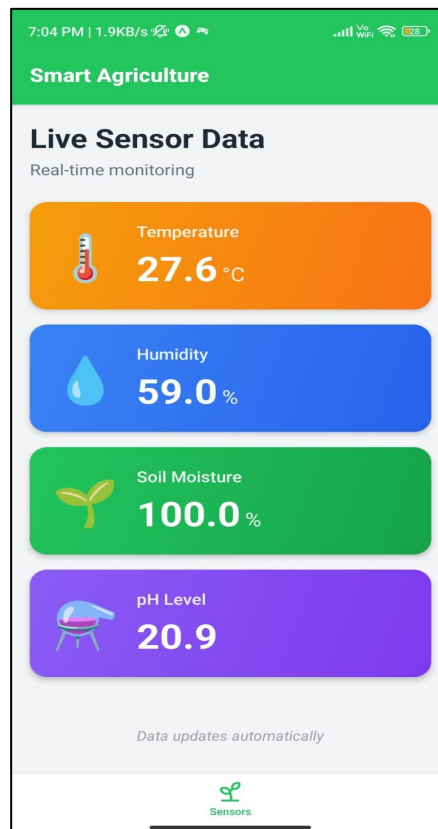
```

Observations / Result:

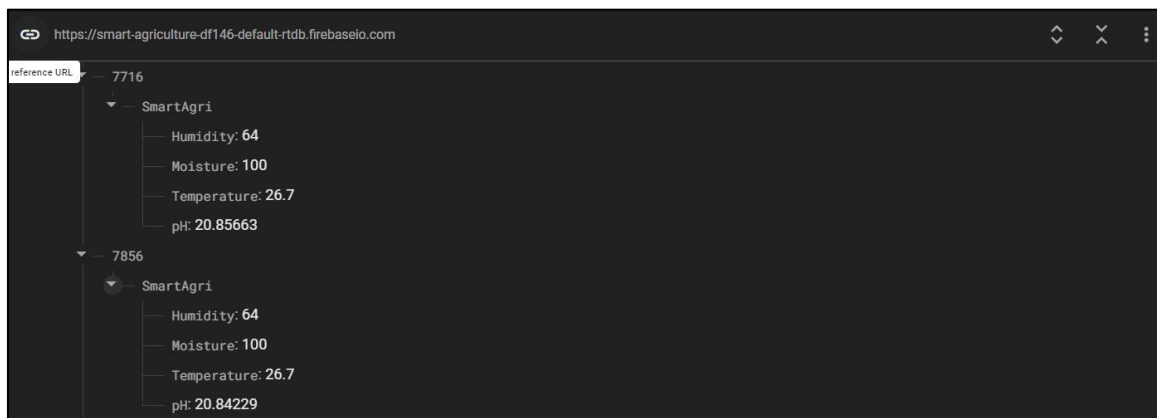
- Serial Monitor Output:-



- Mobile App UI:-



- Cloud Data Logs:-



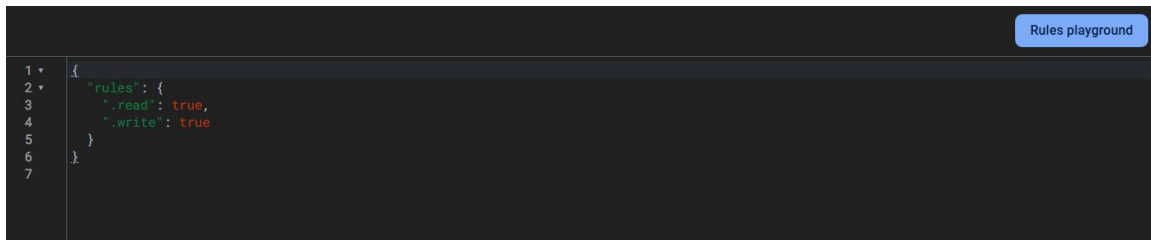
Troubleshooting & Debugging:

1. Wi-Fi Connection Timeout

- Problem: The ESP8266 repeatedly failed to connect to Wi-Fi.
- Cause: Incorrect SSID/password or poor Wi-Fi signal.
- Solution: Verified credentials, moved the module closer to the router, and implemented a reconnection loop in code using `WiFi.status()` checks.

2. Data Not Updating in Firebase

- Problem: Sensor values did not appear in Firebase Database.
- Cause: Firebase database security rules restricted write access.
- Solution: Temporarily set database rules to test mode:



```
1 {
2   "rules": {
3     ".read": true,
4     ".write": true
5   }
6 }
7
```

and confirmed successful data uploads.

3. Inconsistent Sensor Upload Rates

- Problem: Data was sent irregularly or stopped updating.
- Cause: Insufficient delay and unstable loop cycle.
- Solution: Added delay(2000) in the loop for stable upload intervals.

4. Serial Monitor Displaying “NaN” Values

- Problem: DHT11 sensor sometimes returned “NaN” (Not a Number).
- Cause: Timing issue or incorrect initialization of DHT library.
- Solution: Added dht.begin() in setup() and inserted a 2-second delay before reading.

Conclusion:

In this practical, we established successful cloud connectivity between the ESP8266 microcontroller and Firebase Realtime Database.

The microcontroller collected sensor data and uploaded it to the cloud, enabling remote monitoring of environmental parameters.

We learned:

- How to configure and authenticate Firebase for IoT data storage.
- How real-time databases maintain continuous synchronization between device and cloud.
- How cloud connectivity enables remote accessibility of IoT data anytime, anywhere.

This practical demonstrates the backbone of modern IoT systems — real-time cloud integration — which is critical for applications like smart farming, environmental sensing, and home automation.